Sorting Trays After New Labels

Unit 3 · Topic 3.7 · THE PROBLEM

A cafeteria served 5,000 September trays after new sorting-bin labels were installed. The director claims at least 60% are sorted correctly. An SRS of 250 trays finds 136 correct; the club tests at $\alpha = 0.05$.

Imani: "The p-value is 0.0354, so reject H_0. The labels caused sorting below 60 percent for every month."

Drew: "The 5.6-point gap may be too small to matter, so it is not significant. Accept the director's claim."

The test conditions have been checked: random sample; $250 < 500$; null-expected counts 150 and 100.

September study

Source	Trays	Correct	α
September	5,000	claim: $\geq 60\%$	–
SRS	250	136	0.05

FIRST TAKE — one sentence before you work the parts

Can these September trays tell us what caused the result in every month? Give one reason from the study.

DOK 1 **(a)** Define p and state the hypotheses for checking whether correct sorting is below 60%.

DOK 2 **(b)** Calculate $\widehat{p}$, the one-proportion z statistic, and the lower-tail p-value. Use the verified conditions above.

DOK 3 **(c)** In 4–5 sentences, correct both reports. Compare the p-value with α, separate statistical significance from practical importance, and limit the conclusion to the people or items studied and what the design can show about cause.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.